



Bus@Belimo

Bus Basics B1


BELIMO

Final Section

Summary

 **BELIMO**
EDUCATION

Agenda Thursday

Bus@Belimo **Bus Basics B1**



KNX

**IT - Basics
IoT**

Lunch@Belgusto

**"Multi-Bus-Building"
Summary**

Test

**Review
Feedback**

Competitors

Most important differentiation points

- Wide range vs. partial coverage
 - Keyword: Danfoss NovoCon
- Additional features
 - Conversion of sensor signal
 - Local forced control
 - Hybrid mode
 - Calculation of the combined heating/cooling sequence (6-way)
- Parameterisation
 - Commissioning tools
 - Commissioning via bus
- Wiring
 - Daisy chain, screw terminal, cabling system
- Price
 - ⇒ Scope and mapping of data points often identical

Competitor Examples



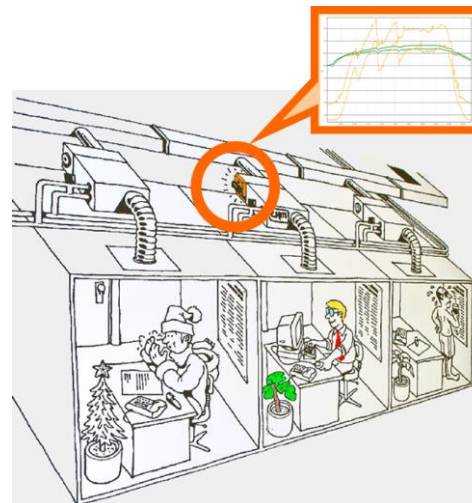
Digitalisation - Why?

Transparence

Commissioning, Optimisation of operation, Maintenance



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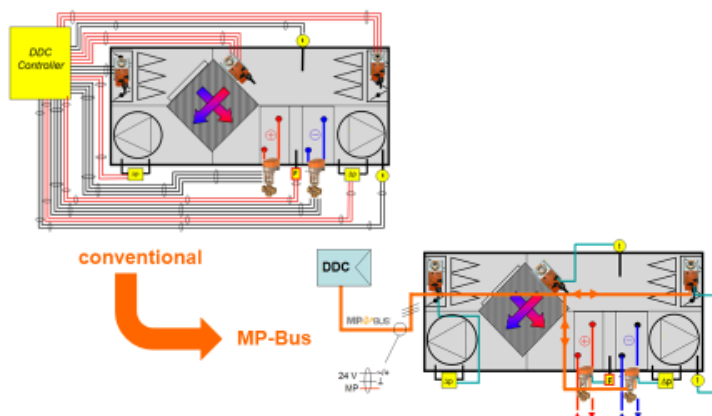
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Building Technology
Energy Monitoring & Optimisation

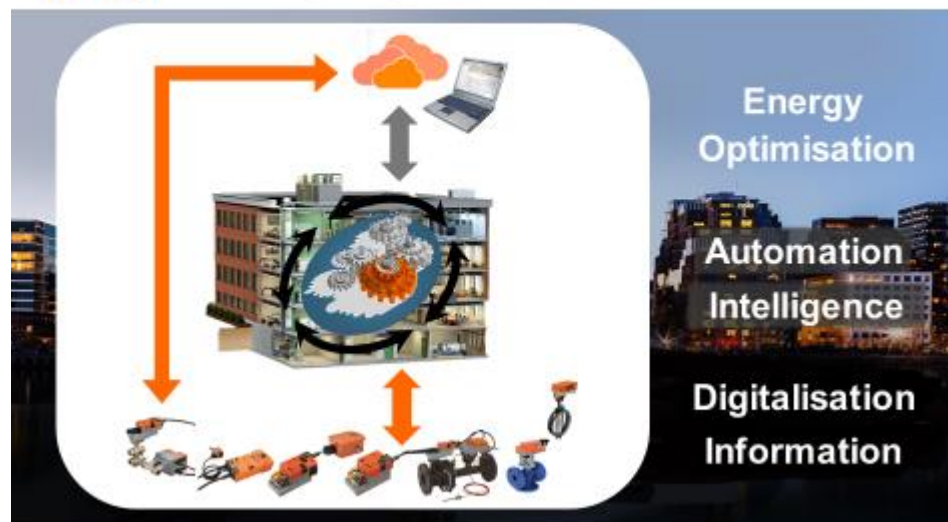
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Example conventional ⇔ MP-Bus

Wiring of Air Handling Unit AHU



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Bus Basics B1 - Building Automation, 2019

11/71

What does every bus protocol need?



- Defined transmitters and receivers:

① Devices / Nodes



- Defined transmission path:



② Wiring and topology

- Unique addresses:

③ Addressing



- Defined data representation / packaging:

④ Data points, operating values

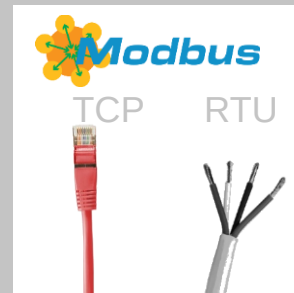
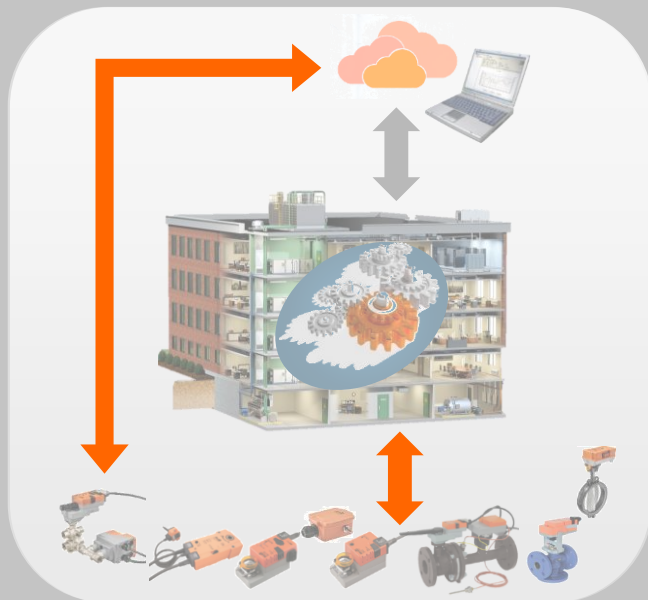


- Defined rules for transmission:

⑤ communication principles

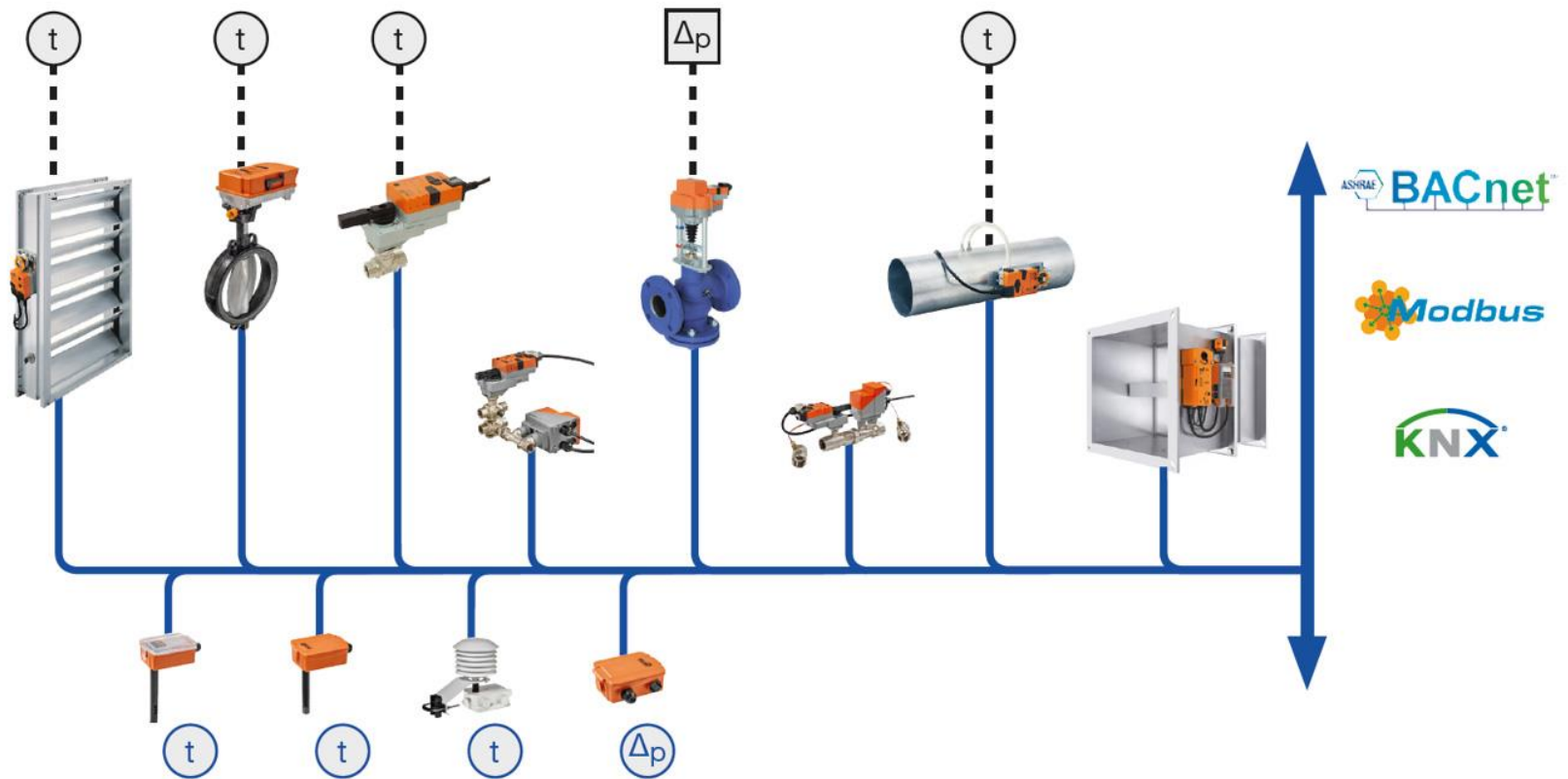


Bus@Belimo in the Building

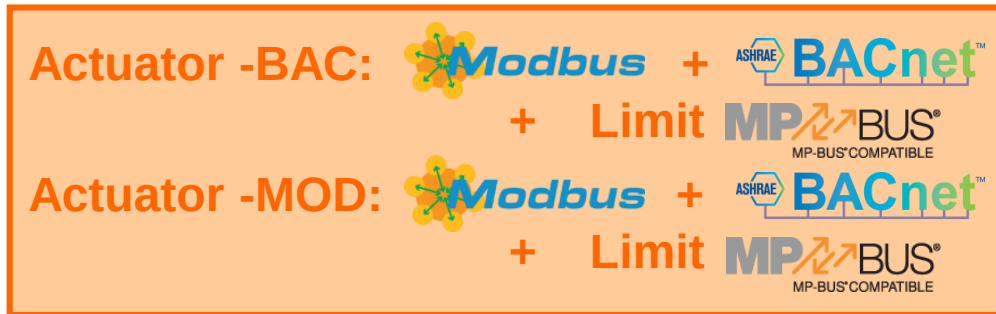


For direct system integration

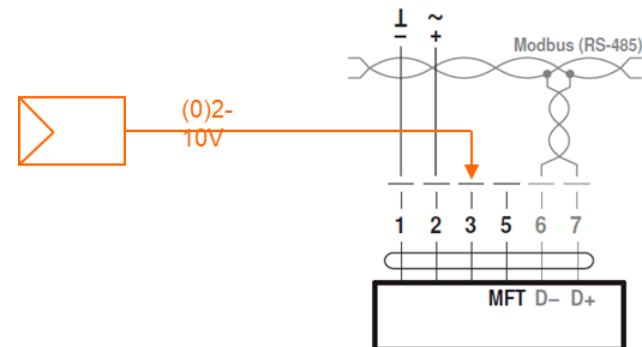
BACnet - Modbus - KNX



Portfolio Actuator Modbus / BACnet

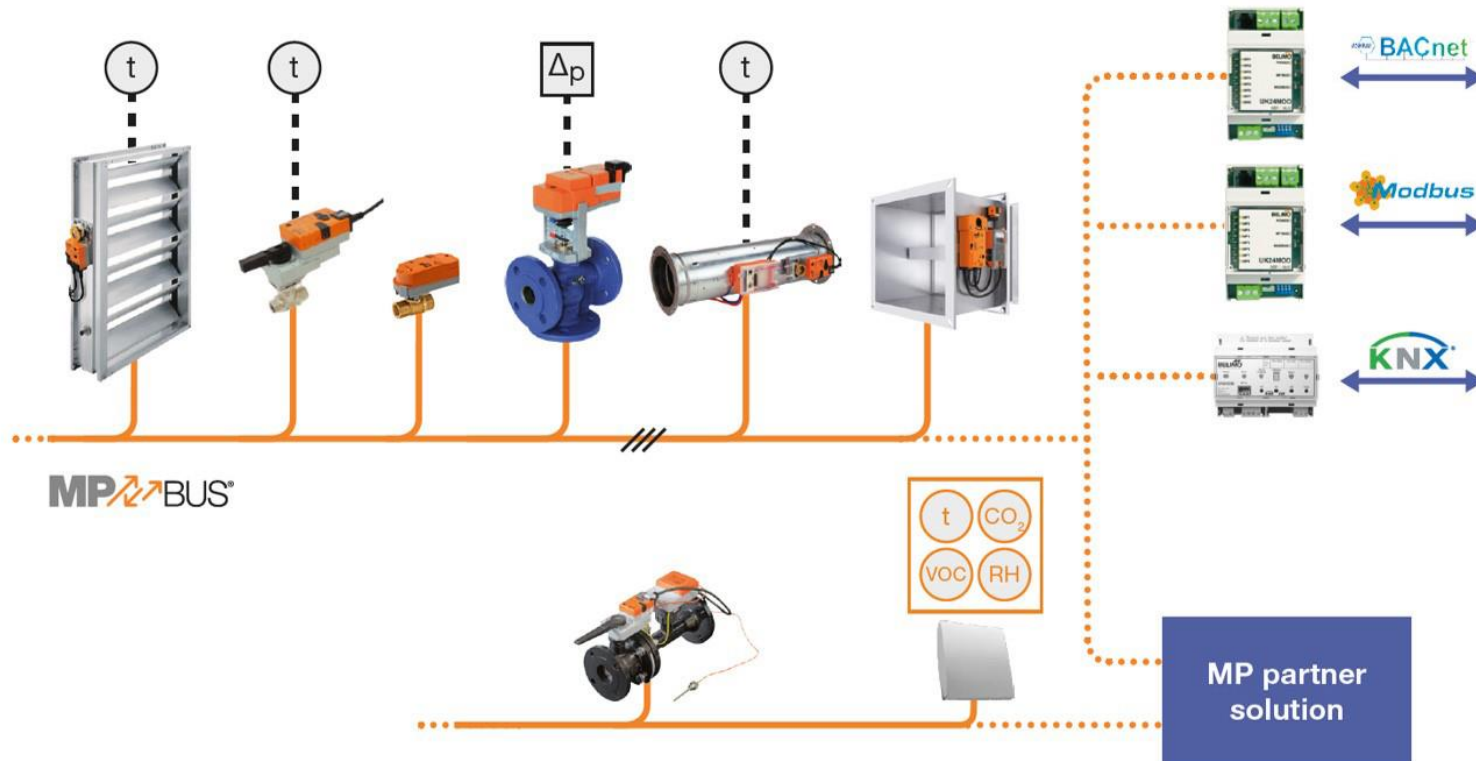


- Backwards compatible
- Hybrid mode for Modbus and BACnet
 - Control Analog
 - Monitoring and override control Bus
- MP Bus supported with Limitation:
 - no COM LED for MP
 - no hybrid mode



Belimo MP-Bus - Integration via Gateway

MP - UK24xxx - BACnet / Modbus / KNX



Attention: Not all actuators / sensors are supported by the gateways!

Bus@Belimo Portfolio in Detail

Bus Brochure and Device Selection Tool



www.belimo.eu/busdevice

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Produkte & Lösungen | Dokumentation | News

Lüftungsanwendungen
 Wasseranwendungen
 Sensoren
 Sicherheitsanwendungen
 Raum- und Systemanwendungen (inkl. VAV und Raumregler)
 Bus- und Systemintegration
 Referenzberichte
 PDF-Datenblätter senden

Adobe Reader herunterladen

Bus- und Systemintegration

- Belimo MP-Bus®
- LonWorks®
- Modbus RTU
- KNX
- BACnet
- Profibus DP
- Belimo Cloud
- MFT-Parametrier-T
- Zubehör (Buskommunikation)
- Abgelöste Produkte
- Weitere Broschüren
- Referenzberichte

Belimo MP-Bus®

- Info-Broschüre: t
- Einführungsanleitung

Bus solutions from Belimo.
The best connection.

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Device selection tool

Filter bus devices

Direct integration

- ☐ MP-Bus
☒ Modbus
 - ☒ RTU
 - ☐ TCP

- ☐ BACnet
☐ LON
☐ KNX
☐ Cloud

Sensor connection

- ☐ Active (0-10 V)
☐ Active (0-32 V)
☐ Passive (e.g. PT1000)
☐ Switching contact

Type

- ☒ Actuator
☐ Field device
☐ Sensor

Reset

MP-Bus with gateways

- ☐ via UK24MOD
☐ via UK24BAC
☐ via UK24LON
☐ via UK24EIB

Application

- ☐ Air
☐ Safety
☐ VAV
☐ Water

Fail-safe function

- ☐ with
☐ without

66 suitable devices

AVK24A-MOD



Globe v
AC/DC
Commu
MP-Bus
Running
Running
IP54, M

CQ24A-BAC



Rotary a
AC/DC
Commu
Running
IP40, M
Connec

EP015R+MOD



Electr. 2
Internal
AC/DC
Commu
MP-Bus
Adjusta
IP54, M
locked

EP015R-R6+BAC



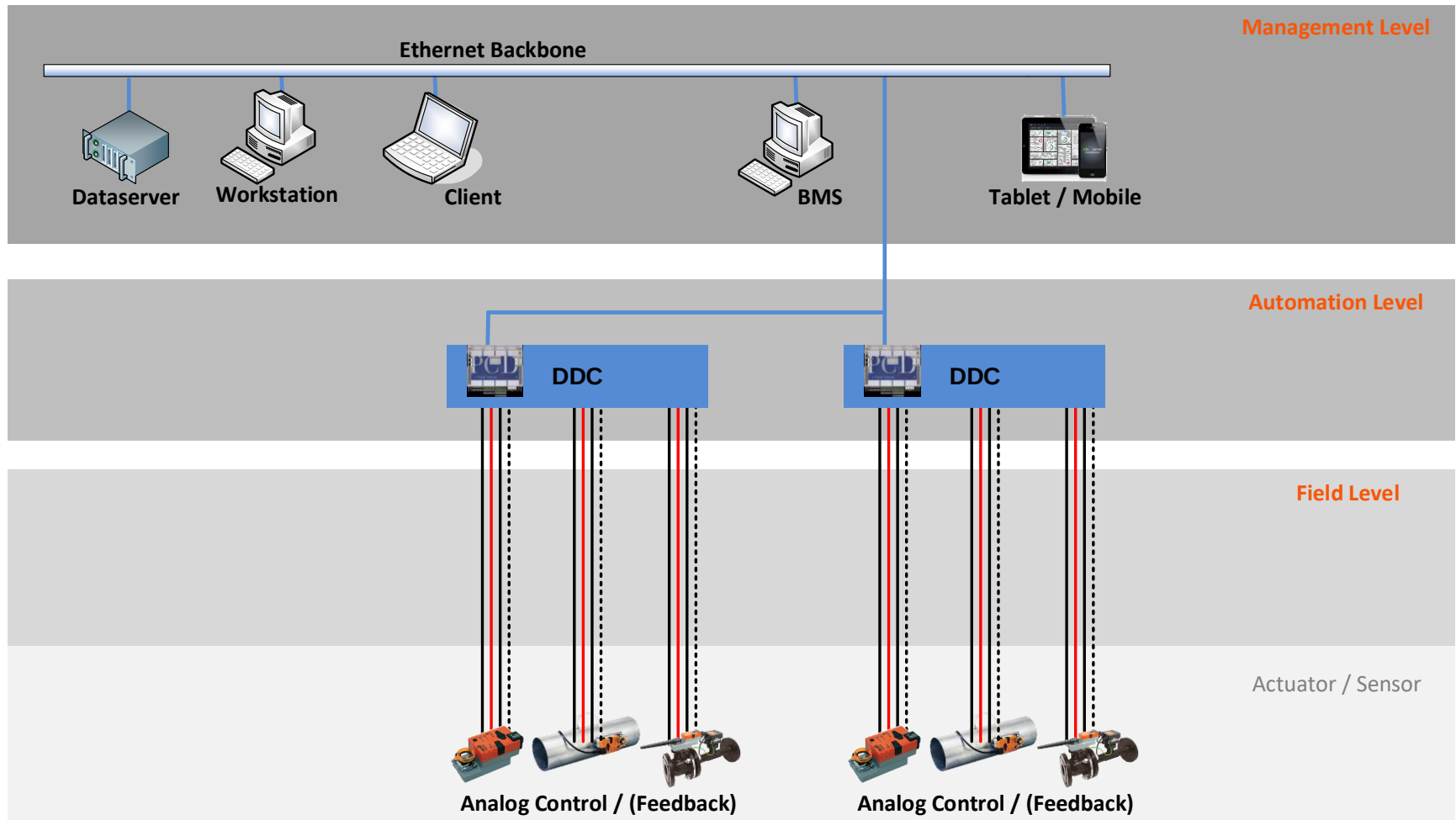
Electr. 6
Internal
AC/DC
Commu

Booklet Summary



Building Topology **YESTERDAY** ⇒ **TODAY**

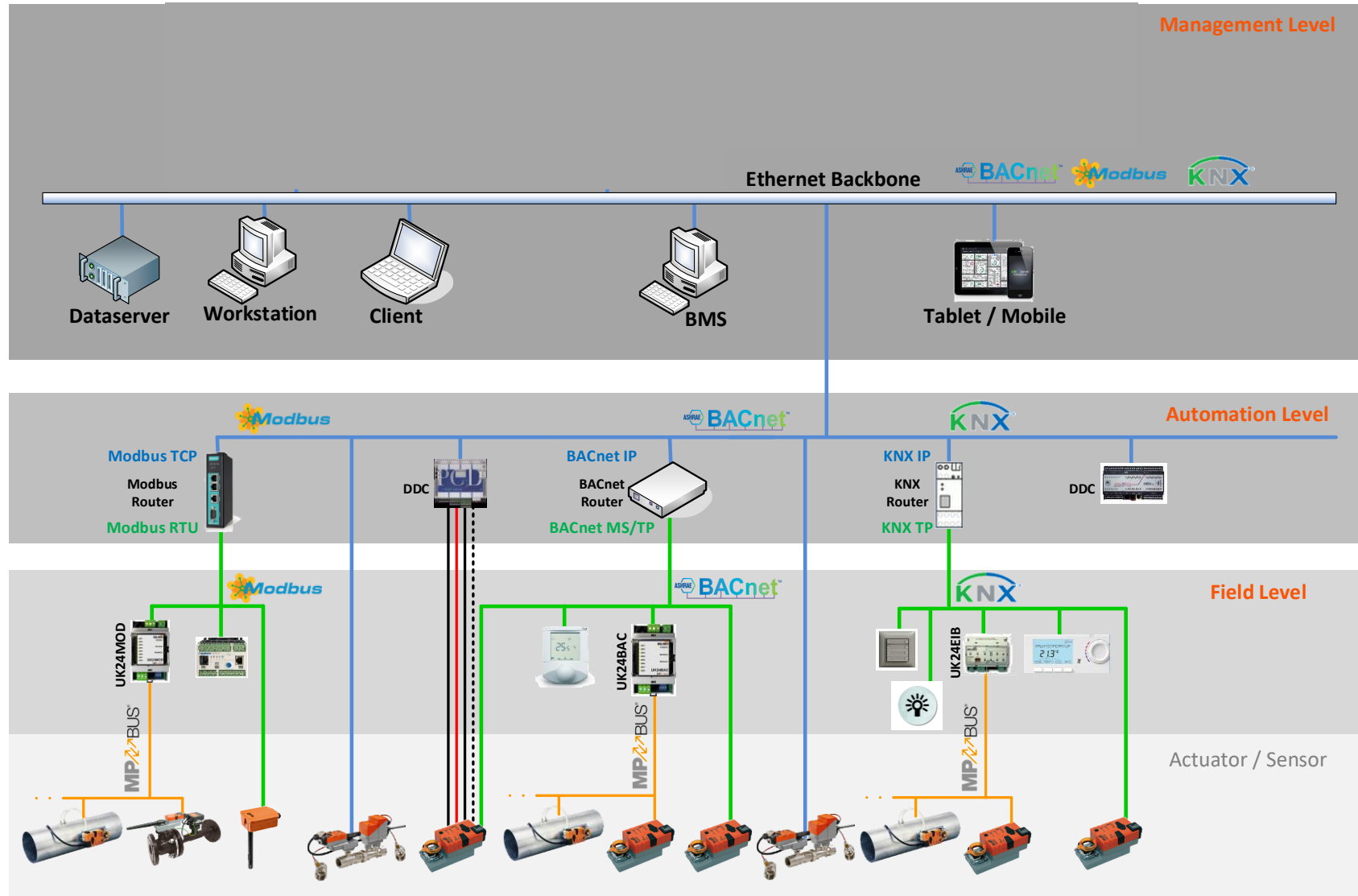
Field Level: conventional – analog control



Building Topology

TODAY ⇒ ?

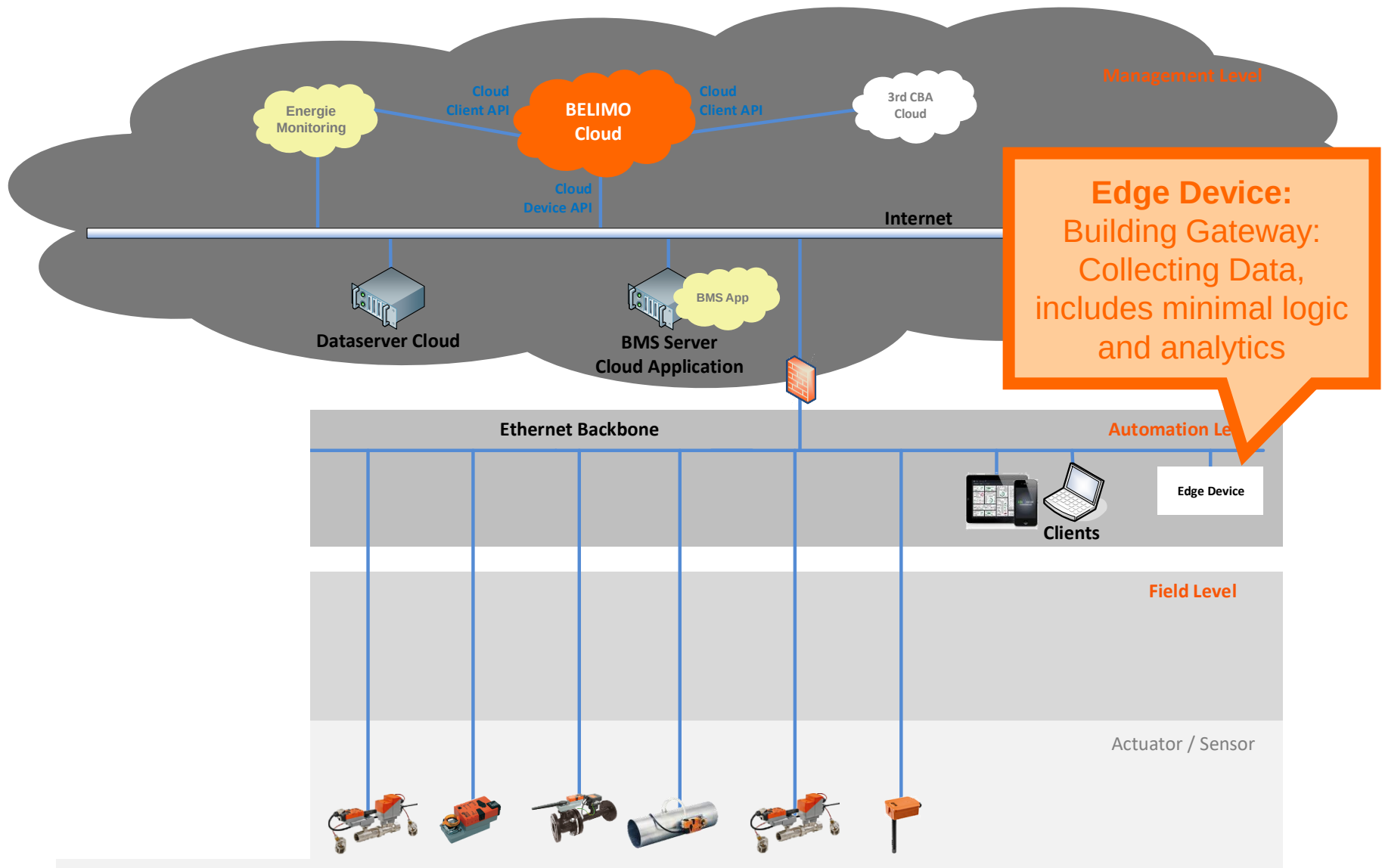
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Building Topology

FUTURE

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Booklet Summary



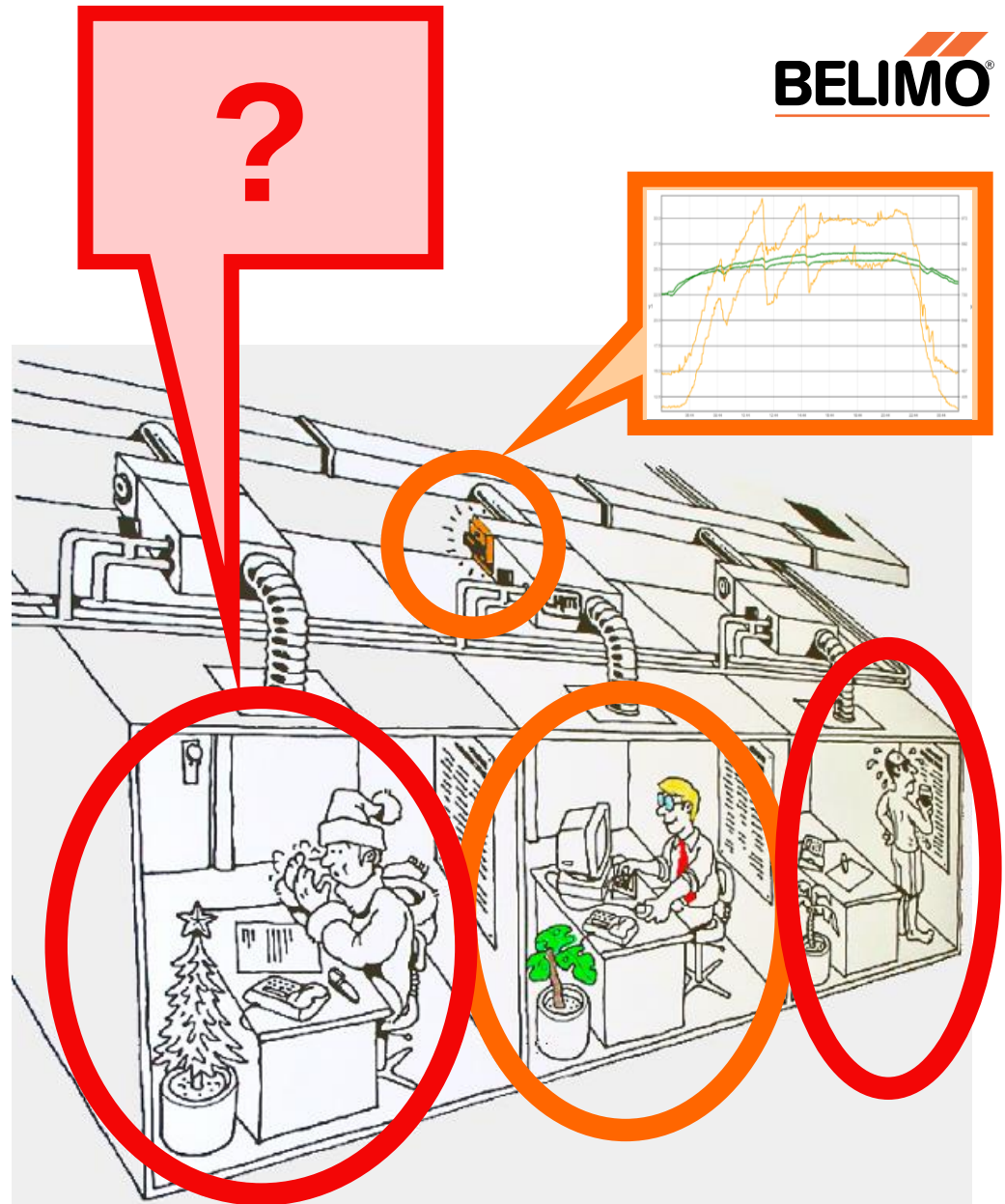
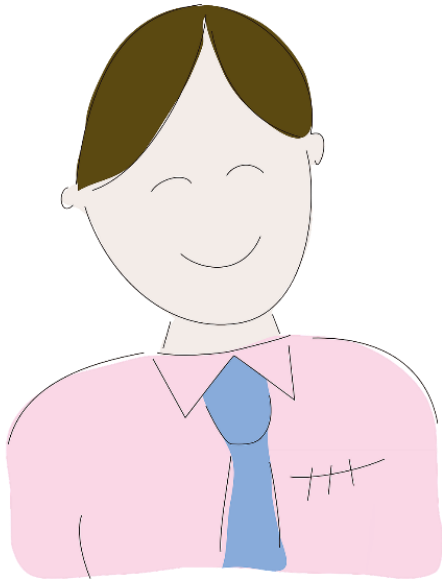
Any questions?



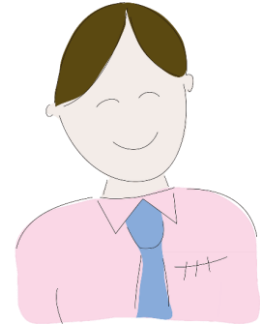
- Fill in the Test on your own
- Solution in plenum

Comfort Climate and Safety

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Feedback and Expectations met?



Online Feedback

The screenshot shows a web form for BELIMO. At the top, the BELIMO logo is in the upper left, and the title 'News Training EN' is in an orange bar. Below this is a 'General' section header. The first question, marked with an asterisk, is '1. Contact details' with a speech bubble icon. It contains two text input fields labeled 'Name' and 'Country'. The second question, also marked with an asterisk, is '2. Attended training' with a speech bubble icon. It has two radio button options: '12./13.4.2018' and '17./18.4.2018'. The third question, marked with an asterisk, is '3. How much did you like the News Training 2018?' with a speech bubble icon. It features a five-star rating scale. The first star is labeled 'Not at all' and the fifth star is labeled 'Very much'. Below the stars is an orange 'Next' button. At the bottom, it says 'Powered by SurveyMonkey' with the SurveyMonkey logo and a link 'See how easy it is to create a survey.'



Have a safe journey home!





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